



# DAVID CAICEDO SAMBONI

MECHATRONICS ENGINEER  
WITH A SPECIALIZATION IN AI

## CONTACT

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 @Deivs117

## TECHNICAL SKILLS

- Edge AI & Embedded Vision
- Robotics & Autonomous Systems (ROS)
- Firmware Development (C/C++)
- Control Systems Engineering (PID)
- Industrial IoT & Telemetry (MQTT)
- Advanced CAD & Structural
- Cloud Infrastructure (Azure)

## PROGRAMMING LANGUAGES

|     |            |        |
|-----|------------|--------|
| C++ | JavaScript | Bash   |
| C   | HTML/CSS   | SQL    |
| C#  | TeX        | Python |

## LANGUAGE

Spanish - Native      English - B2

## BUSINESS

Lead the entire CNC  
subtractive  
manufacturing  
workflow

 @fluxsolutionscali



## ABOUT ME

Mechatronics Engineer specializing in hardware, software, and data. Lead developer of end-to-end technological solutions, integrating robotics, Cloud architecture, and Artificial Intelligence to build smart industrial processes and scalable IoT networks.

## EXPERIENCE

### 2026 Lead Mechatronics Engineer

- Designed and implemented an Azure serverless backend for real-time industrial telemetry acquisition and predictive maintenance strategies.
- Deployed neural networks (CNN/MLP) via Edge Impulse on ESP32 microcontrollers for autonomous PCB classification and vibration anomaly detection.
- Developed C++ firmware utilizing secure MQTTS protocols, ensuring multi-node telemetry synchronization and ultra-low-latency actuator control.
- Created automated Bash pipelines to accelerate infrastructure deployment, successfully integrating a real-time Telegram alert system with a 99% success rate.

### 2025 AI-Driven Children's Emotion Recognition System for Mental Health Records

- I designed a multi-stage inference pipeline that automates real-time demographic segmentation and emotional state classification to support behavioral monitoring in vulnerable pediatric populations.
- I developed custom deep learning classifiers that outperformed generalized open-source models.
- I conducted rigorous benchmarking of edge cases, driving future cycles of MLOps and data engineering by identifying systemic imbalances in the datasets as the primary bottleneck for model scalability.



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SOFT SKILLS

- Cross-Functional Leadership
- Multidisciplinary Collaboration
- Research-Driven Problem Solving
- Technical Adaptability
- Strategic Communication
- Builder Mindset
- Analytical Critical Thinking
- Ownership & Initiative

REGULATORY COMPLIANCE

- ISO 9001
- ISO 10218
- ISO/TS 15066
- ISO 13482
- ISO/IEC 42001
- ISO 13849-1
- IEC 62443

WEBSITE

https://deivs117.github.io/portfolio/

BUSINESS

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2025 SALLI: Salamander Autonomous Locomotion on Land Infrastructure

- I led the development of the SALLI platform from initial conceptualization through technical deployment, successfully coordinating a multidisciplinary team in accordance with ISO 9001 quality standards and planning its future iterations.
- I designed the system's comprehensive mechanical architecture and carried out structural optimization of the chassis, maximizing the device's operational stability, load distribution, and durability in dynamic environments
- I designed and implemented a robust wireless firmware architecture based on a master-slave model using multiple ESP32 microcontrollers, achieving synchronization of a complex modular drive system and real-time telemetry processing.
- I optimized and fully deployed a lightweight convolutional neural network directly on constrained hardware (ESP-CAM) for edge computing, enabling autonomous real-time target tracking and intelligent obstacle avoidance capabilities.

2024 Autonomous Hybrid Quadruped-Wheeled Platform for Multiple Biomes.

- I designed and implemented neural networks to control complex hybrid locomotion strategies, ensuring adaptive stability across different terrains.
- I parameterized and conducted rigorous validation tests for multimodal transition states, optimizing the smooth transition between quadrupedal walking, directional driving, and omnidirectional mobility to 95%.
- I developed and deployed the core autonomous control architecture within the Robot Operating System (ROS), coordinating state estimation and executing real-time locomotion loops.

EDUCATION

Universidad Autónoma de Occidente

2026 Mechatronics Engineer

2026 Specialization in AI

2025 NeuroControl Research Group | Robotics & AI  
Researcher